Oxford College of Emory University

Chapter 15: Monetary Policy

Part 1: Money Demand, Interest Rates, and Aggregate Demand

ECON_OX 112 · Principles of Macroeconomics

Guilherme Paiva Pinto

September 2024: A Rate Cut

- On September 18, 2024, the Fed cut its target range by 0.50 percentage point, to 4.75–5.00%.

FOMC, September 18, 2024

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September 2024: A Rate Cut

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- Imagine a bakery that needs a loan for a new oven. A lower borrowing cost could make the purchase worthwhile.
- How can a change in an overnight bank rate affect the bakery’s loan?

FOMC, September 18, 2024

Roadmap

- Why people hold money
- How money supply and money demand affect interest rates
- How monetary policy affects spending and aggregate demand

15.1 Demand for Money

- Understand and explain why people hold money and what changes the amount they want.

Money or Bonds

- A household has $1,000 and needs $400 for bills next month.
- Money earns 0%. The bond pays interest but cannot be cashed in early in this exercise.
- Answer P1–P4: how much should the household keep available for payments?

The Opportunity Cost of Holding Money

Keep $600 for one year	Ending amount	Interest earned
Money at 0%	$600	$0
Bond at 1%	$606	$6
Bond at 6%	$636	$36

Rate of return = (ending amount − starting amount) ÷ starting amount

- For the 6% bond: $36 ÷ $600 = 6% per year.

SEC Investor.gov: bonds · SEC Investor.gov: certificates of deposit

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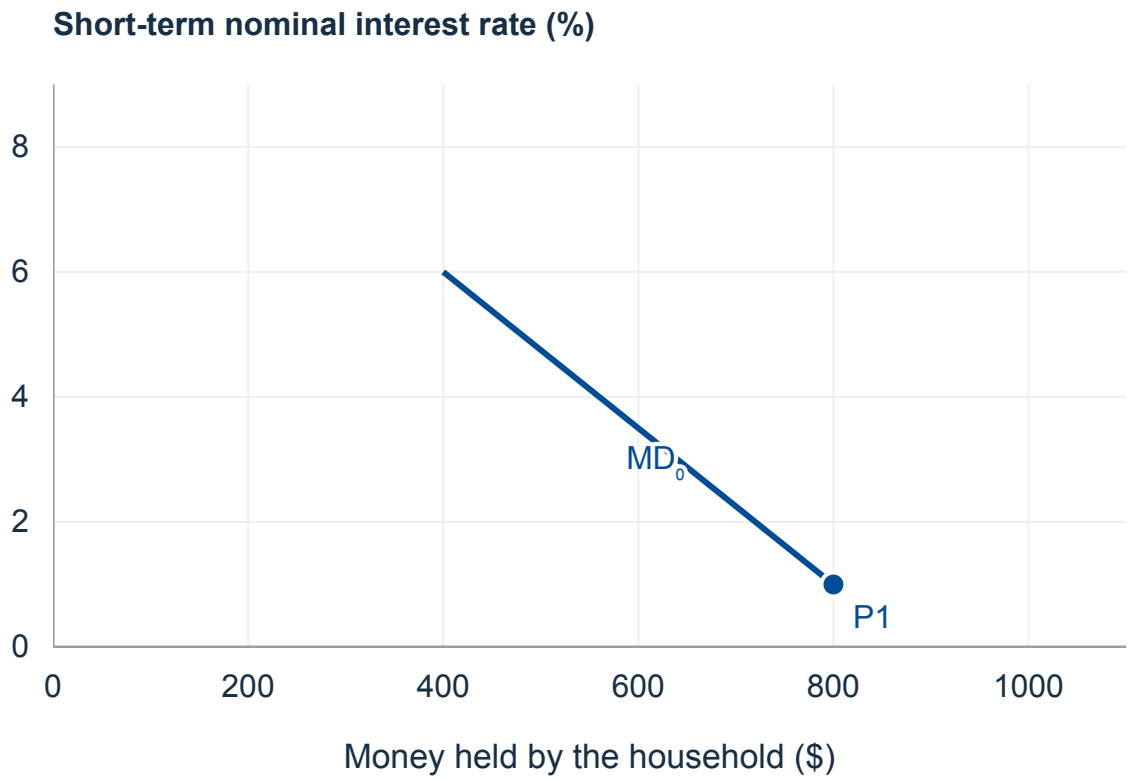
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- Keeping $600 in money means giving up that interest.
- When the bond rate rises from 1% to 6%, interest forgone rises from $6 to $36.

SEC Investor.gov: bonds · SEC Investor.gov: certificates of deposit

Money Demand: P1–P3

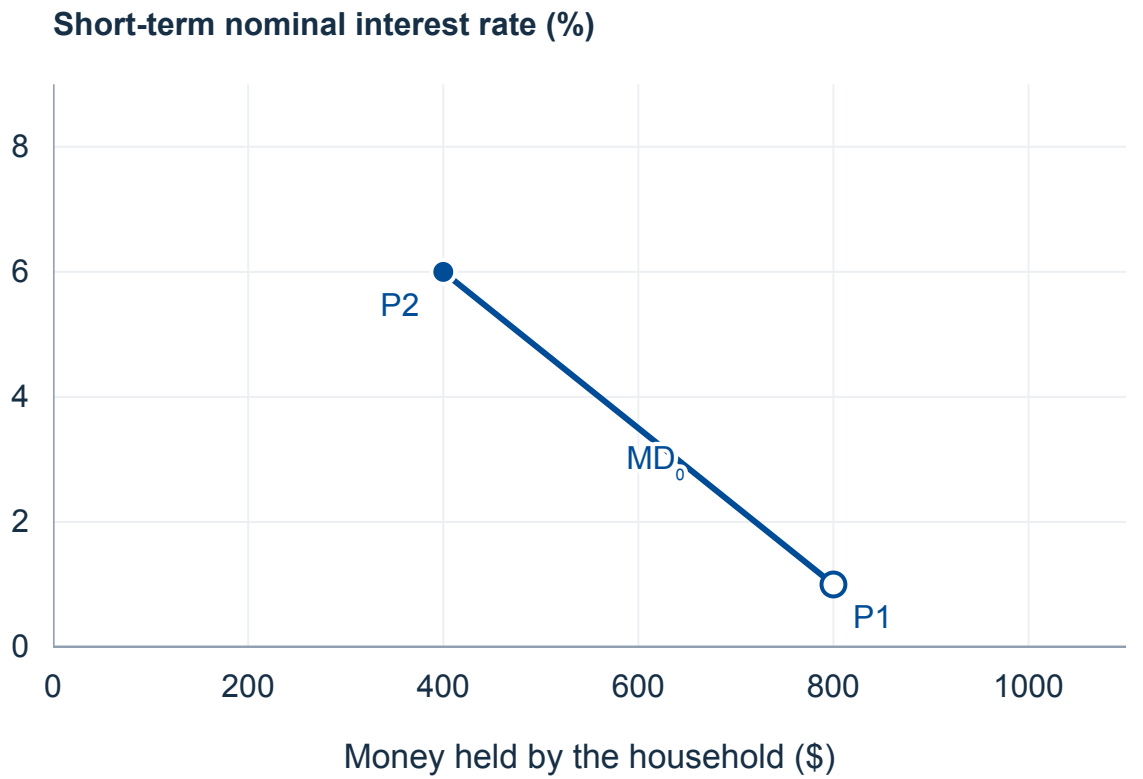


Illustrative choices from P1–P3; not class vote totals.

P3 changes the bills, not the interest rate.

P1: at 1%, the example household keeps $800 for spending.

Money Demand: P1–P3



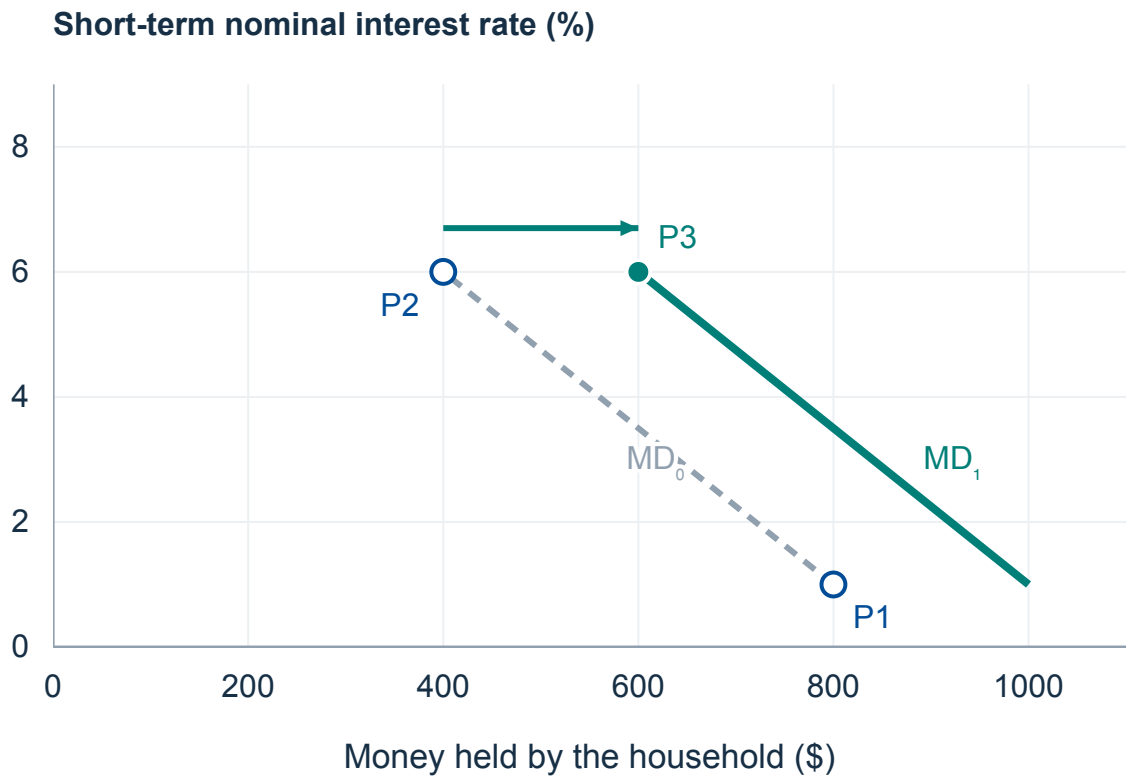
Illustrative choices from P1–P3; not class vote totals.

P3 changes the bills, not the interest rate.

P1: at 1%, the example household keeps $800 for spending.

P2: at 6%, it keeps $400. This is a movement along MD_0.

Money Demand: P1–P3



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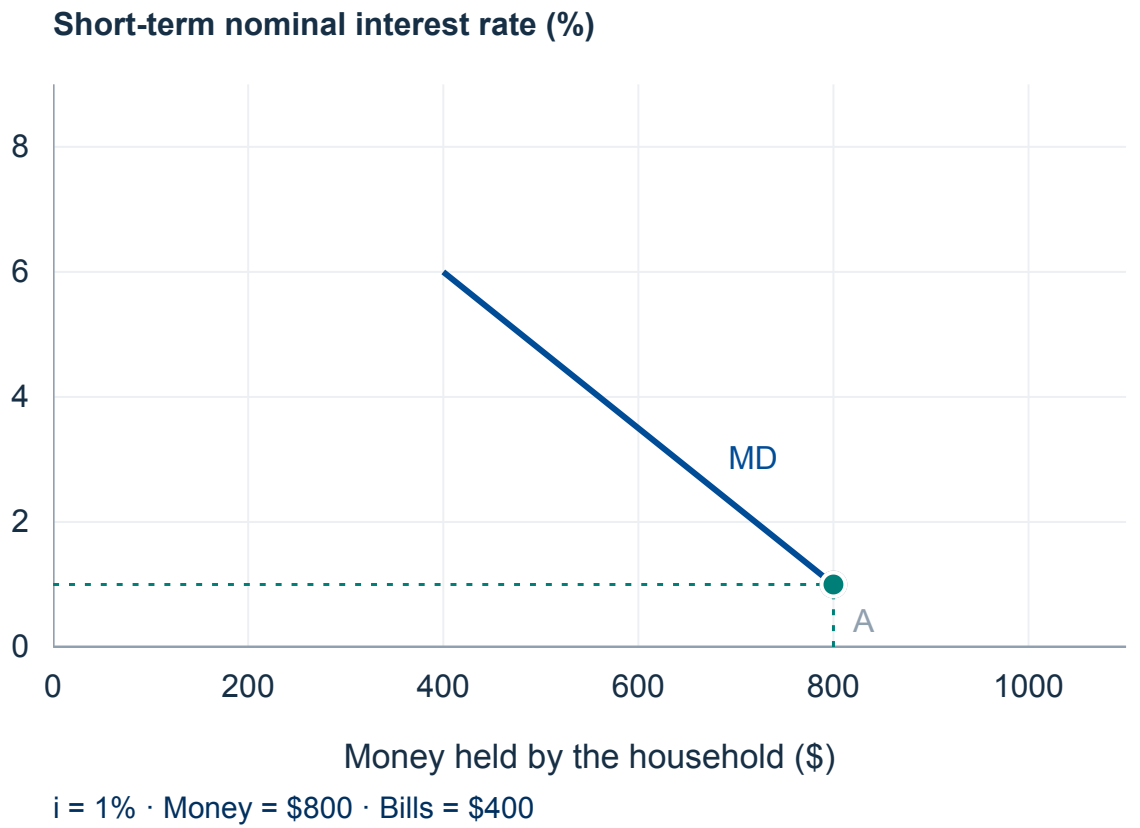
P3 changes the bills, not the interest rate.

P1: at 1%, the example household keeps $800 for spending.

P2: at 6%, it keeps $400. This is a movement along MD_0.

P3: bills rise from $400 to $600, but the rate stays at 6%. MD shifts right to MD_1.

Interest Rates and Money Demand

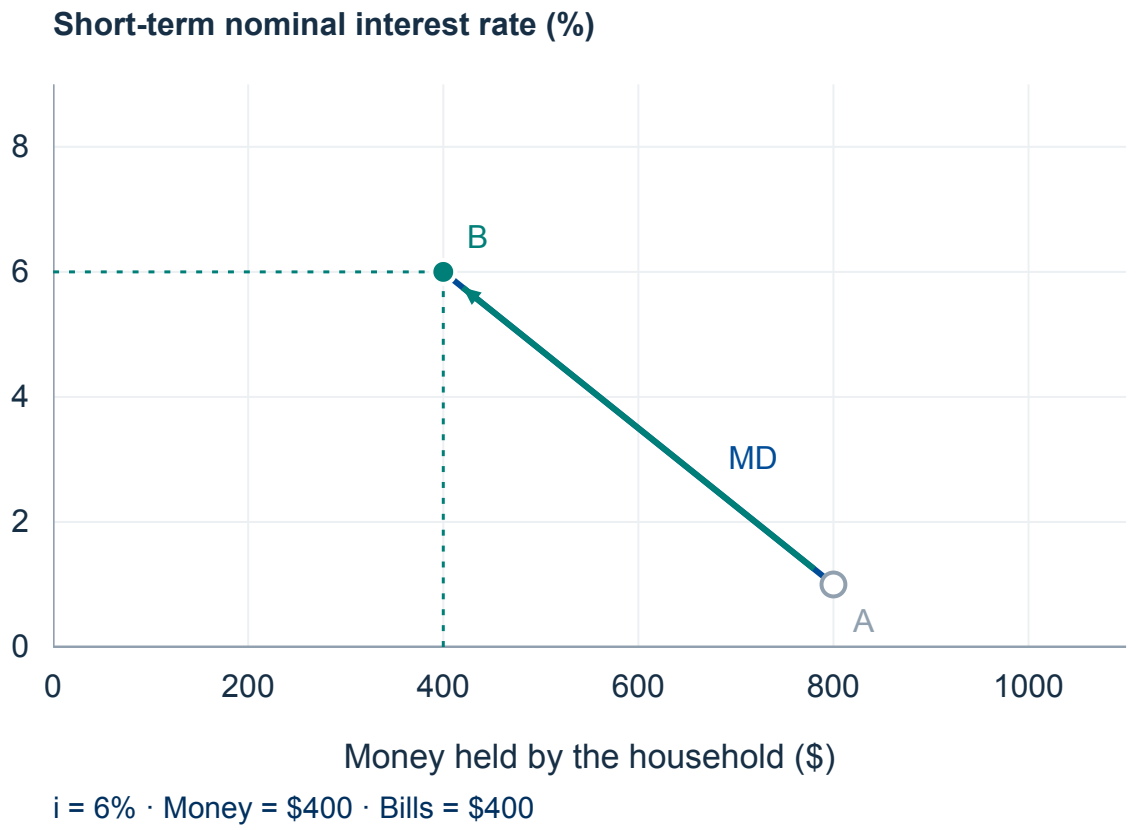


i = 1% · Money = $800 · Bills = $400

The same household: $1,000 total, $400 needed for bills.

At 1%, the household keeps $800 in money.

Interest Rates and Money Demand



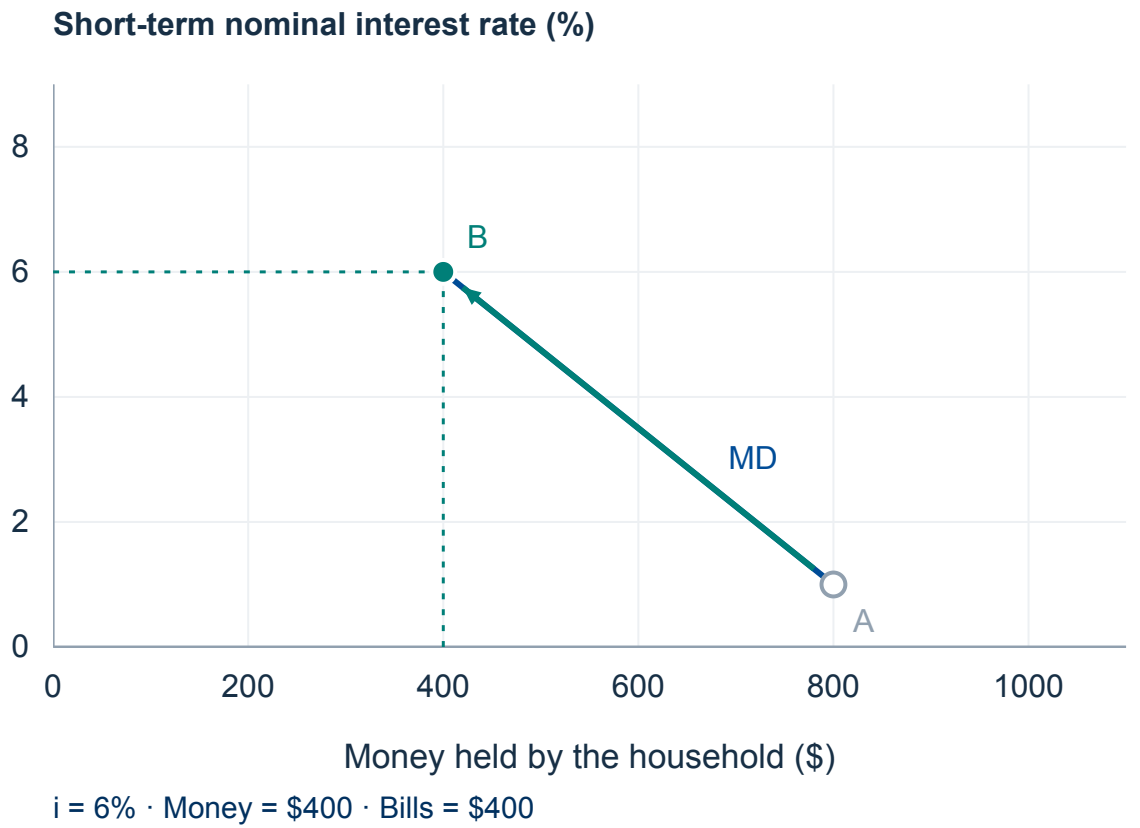
i = 6% · Money = $400 · Bills = $400

The same household: $1,000 total, $400 needed for bills.

At 1%, the household keeps $800 in money.

At 6%, it keeps $400 and puts more into bonds.

Interest Rates and Money Demand



i = 6% · Money = $400 · Bills = $400

The same household: $1,000 total, $400 needed for bills.

At 1%, the household keeps $800 in money.

At 6%, it keeps $400 and puts more into bonds.

Bills still cost $400. Only the rate changed: move along the same money-demand curve.

Shifts in Money Demand

Price level

Real GDP

Credit and banking technology

Banking institutions

Shifts in Money Demand

Price level

Higher prices require more dollars for the same purchases. MD shifts right.

Real GDP

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Shifts in Money Demand

Price level

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More output means more purchases and payments. MD shifts right.

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Shifts in Money Demand

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More output means more purchases and payments. MD shifts right.

Credit and banking technology

Easier credit or faster payments can reduce cash buffers. MD may shift left.

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Shifts in Money Demand

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Higher prices require more dollars for the same purchases. MD shifts right.

Real GDP

More output means more purchases and payments. MD shifts right.

Credit and banking technology

Easier credit or faster payments can reduce cash buffers. MD may shift left.

Banking institutions

Rules that require larger account balances raise money demand. MD shifts right.

Money Demand Checkpoint

1. Interest rates rise; prices and spending stay the same. Movement along MD or a shift?

2. At the same interest rate, the same bills cost more dollars. Movement or shift?

Money Demand Checkpoint

1. Interest rates rise; prices and spending stay the same. Movement along MD or a shift?

> Move up along MD: the quantity of money demanded falls.

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Money Demand Checkpoint

1. Interest rates rise; prices and spending stay the same. Movement along MD or a shift?

> Move up along MD: the quantity of money demanded falls.

2. At the same interest rate, the same bills cost more dollars. Movement or shift?

> MD shifts right: more money is needed for payments.

15.2 Money and Interest Rates

- Explain how money supply and money demand determine the short-term interest rate.

The Federal Funds Rate

River Bank today	Amount
Reserves available	$2,000
Payments due tonight	$2,500
Overnight loan needed	$500

- River Bank can borrow the missing $500 overnight from another bank. The rate it pays is the federal funds rate (FFR).

Federal Reserve Board: policy goals and transmission · Federal Reserve Board: implementation and IORB

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- More reserves reduce banks' need to borrow; fewer reserves increase it. When reserves are scarce, this affects the FFR.

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- The FOMC sets a target range for the FFR.

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- More reserves reduce banks' need to borrow; fewer reserves increase it. When reserves are scarce, this affects the FFR.
- The FOMC sets a target range for the FFR.
- Today the Fed mainly steers this rate by changing the interest it pays on reserve balances.

Federal Reserve Board: policy goals and transmission · Federal Reserve Board: implementation and IORB

A Fed Bond Purchase

Account	The $1,000 purchase
Federal Reserve	Treasury bond +$1,000
River Bank	Reserves: $2,000
Alex	Treasury bond −$1,000

Alex sells a $1,000 Treasury bond to the Fed.

The next graph holds money demand fixed.

Federal Reserve Board: implementation and IORB

A Fed Bond Purchase

Account	The $1,000 purchase
Federal Reserve	Treasury bond +$1,000
River Bank	Reserves: $2,000 + $1,000 → $3,000
Alex	Treasury bond −$1,000

Alex sells a $1,000 Treasury bond to the Fed.

The Fed adds $1,000 to River Bank’s reserve account.

The next graph holds money demand fixed.

Federal Reserve Board: implementation and IORB

A Fed Bond Purchase

Account	The $1,000 purchase
Federal Reserve	Treasury bond +$1,000
River Bank	Reserves: $2,000 + $1,000 → $3,000 Alex’s deposit: +$1,000
Alex	Treasury bond −$1,000 Checking account +$1,000

Alex sells a $1,000 Treasury bond to the Fed.

The Fed adds $1,000 to River Bank’s reserve account.

River Bank adds $1,000 to Alex’s checking account. Alex holds a deposit instead of a bond.

The next graph holds money demand fixed.

Federal Reserve Board: implementation and IORB

A Fed Bond Purchase

Account	The $1,000 purchase
Federal Reserve	Treasury bond +$1,000
River Bank	Reserves: $2,000 + $1,000 → $3,000 Alex’s deposit: +$1,000
Alex	Treasury bond −$1,000 Checking account +$1,000

Overnight loan needed: $500 → $0

Alex sells a $1,000 Treasury bond to the Fed.

The Fed adds $1,000 to River Bank’s reserve account.

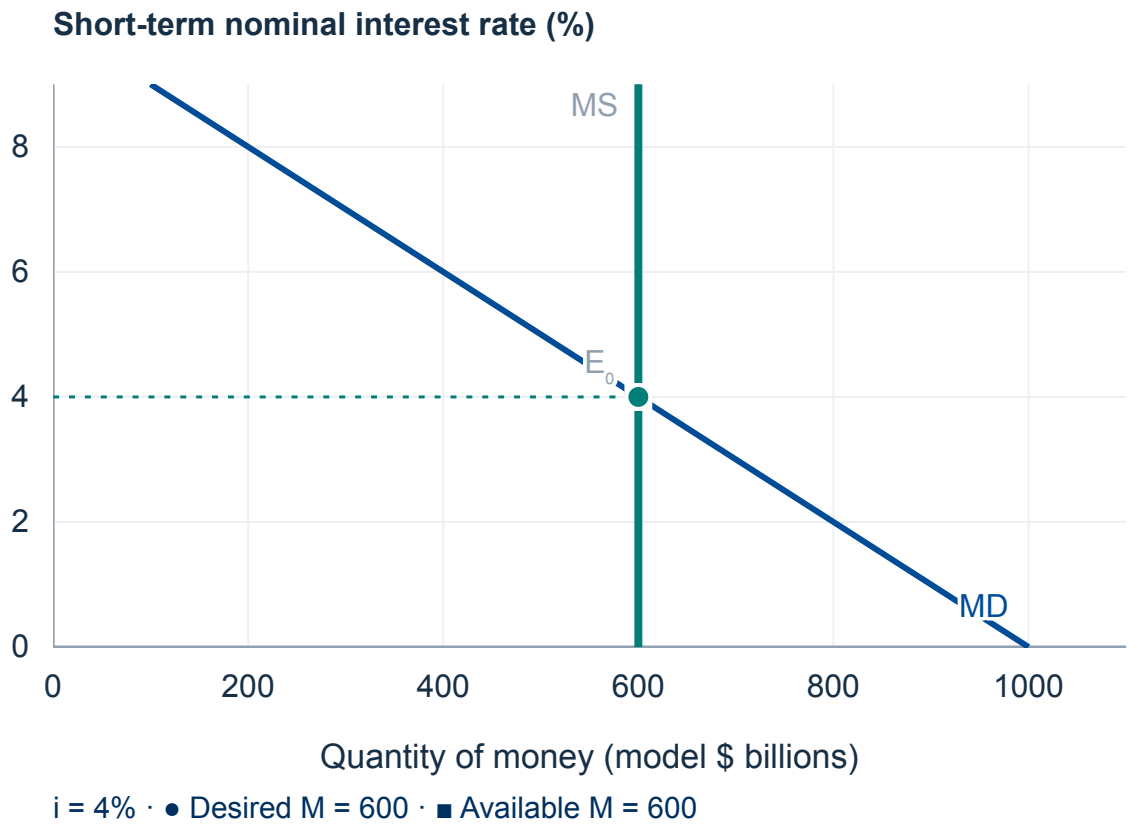
River Bank adds $1,000 to Alex’s checking account. Alex holds a deposit instead of a bond.

Reserves are now $3,000: River Bank can cover its $2,500 payment without borrowing. The new deposit also raises the money supply.

The next graph holds money demand fixed.

Federal Reserve Board: implementation and IORB

Expansionary Monetary Policy



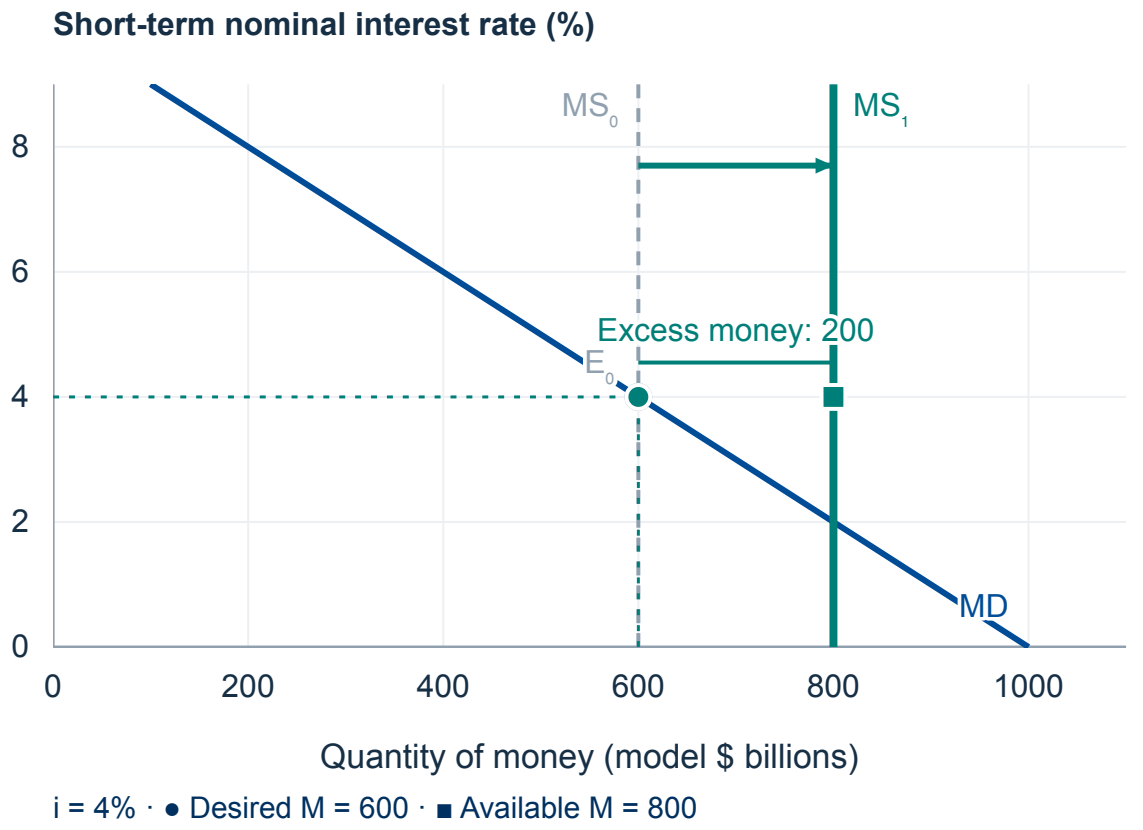
i = 4% · ● Desired M = 600 · ■ Available M = 600

Hold prices, real GDP and the return on money fixed.

Start

Start at M = 600 and i = 4%.

Expansionary Monetary Policy



Hold prices, real GDP and the return on money fixed.

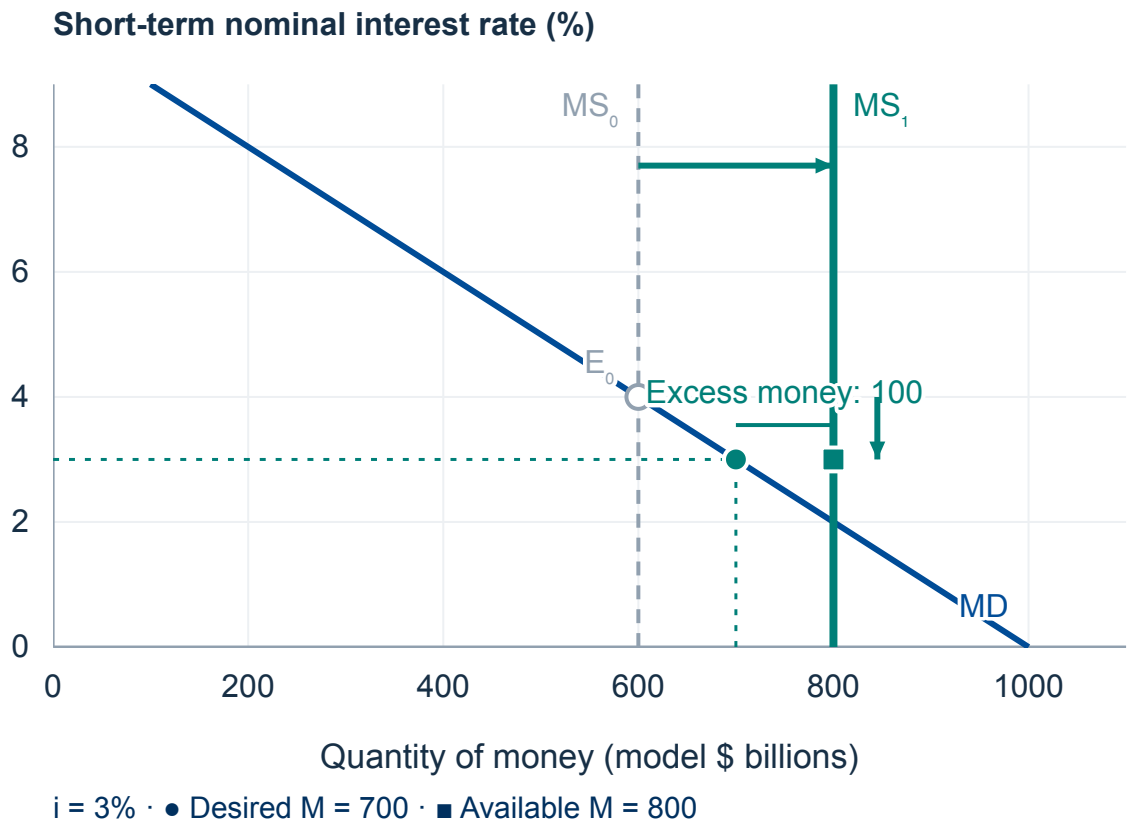
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More money

In this model, the Fed's purchase raises money supply to 800.

Expansionary Monetary Policy



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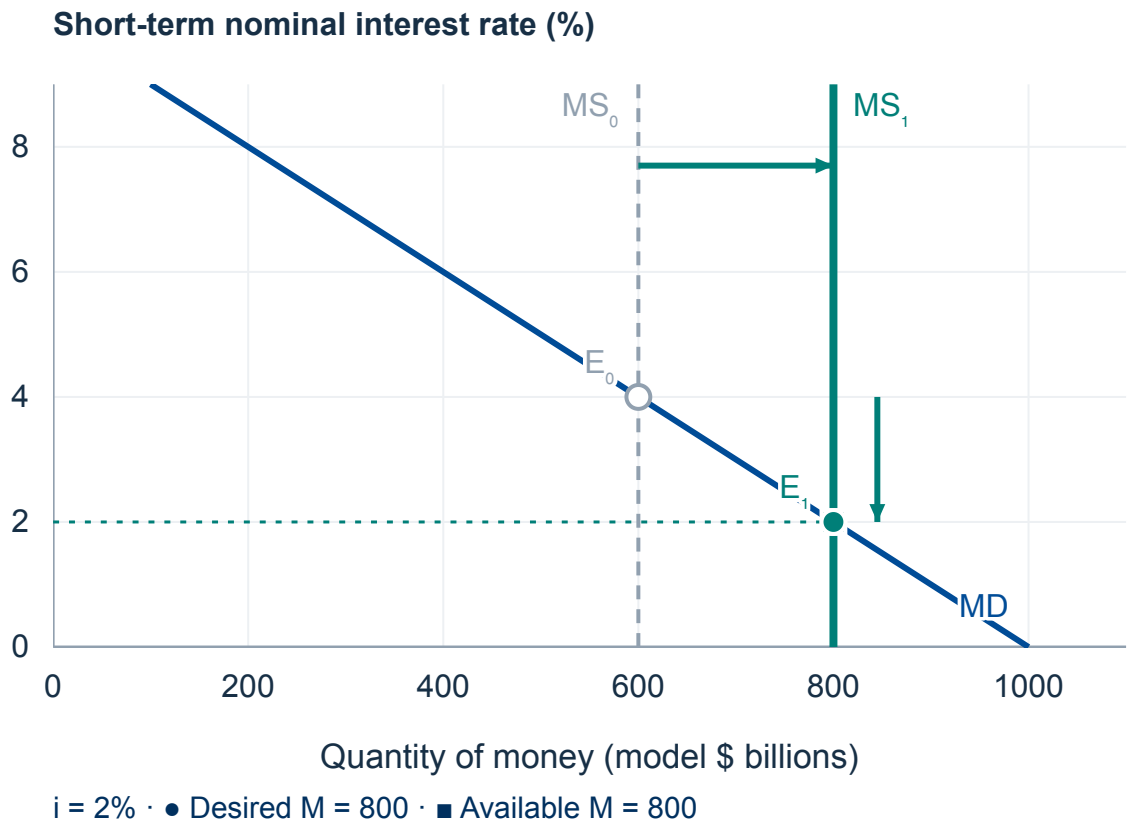
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Bond purchases

At 4%, people hold excess money and use it to buy bonds.

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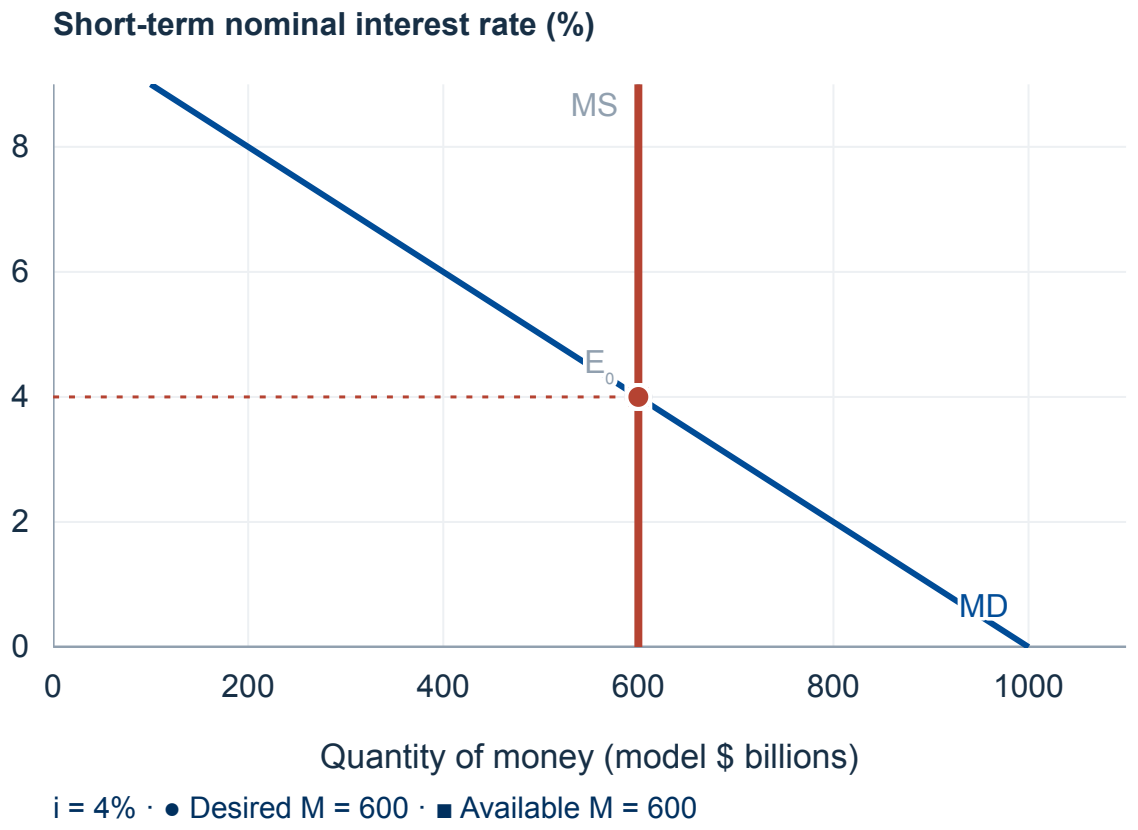
Bond purchases

At 4%, people hold excess money and use it to buy bonds.

Lower interest rate

Bond prices rise. The interest rate falls to 2%.

Contractionary Monetary Policy



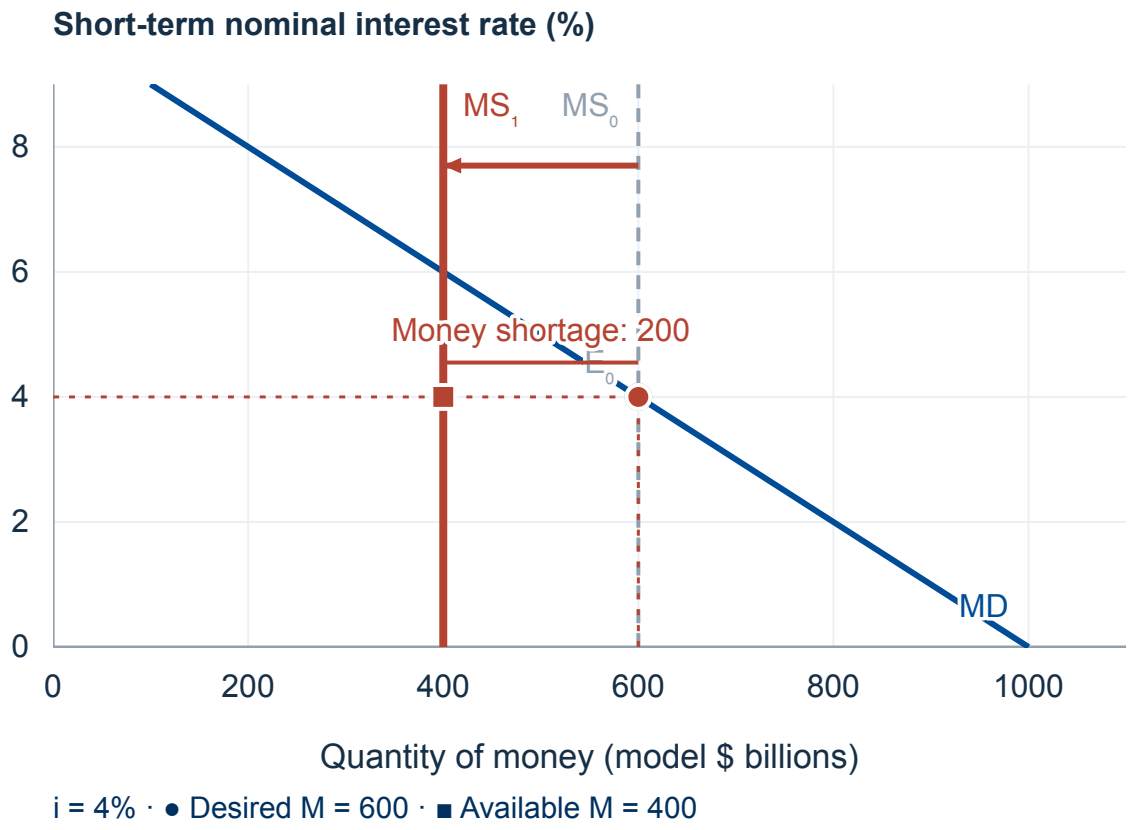
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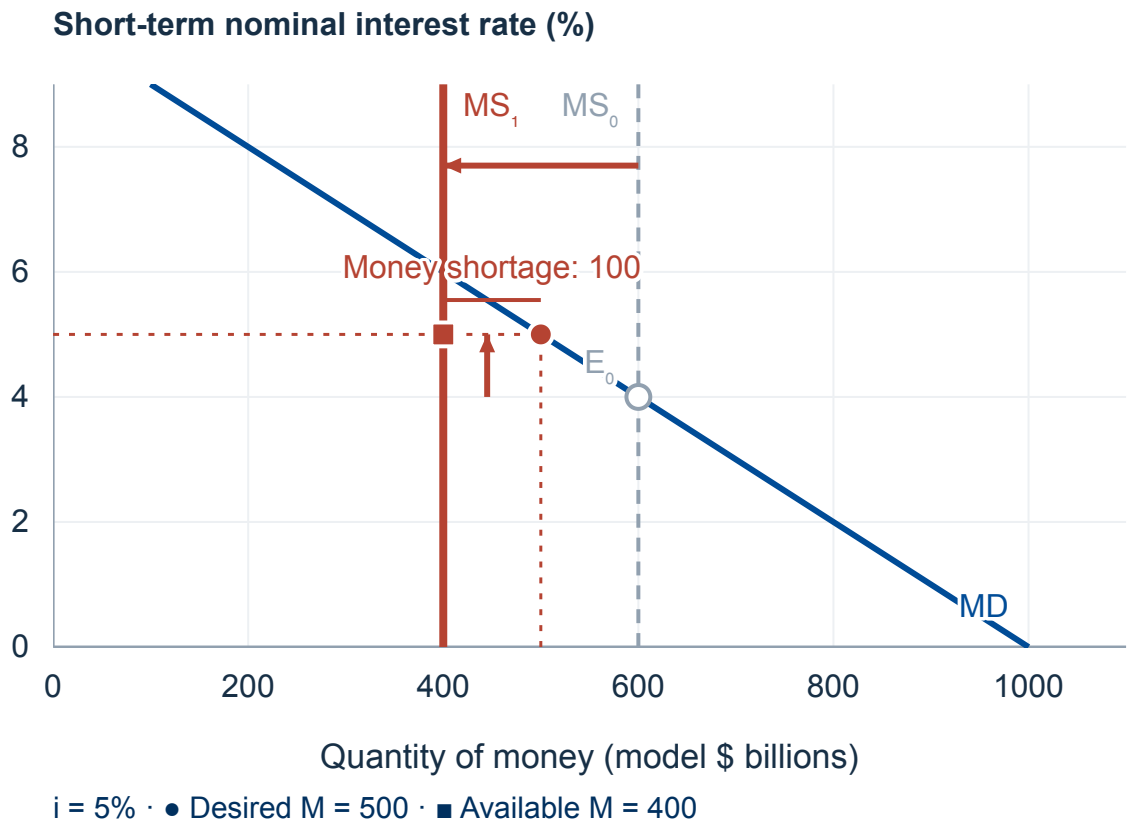
Start

Start at M = 600 and i = 4%.

Less money

In this model, a Fed sale lowers money supply to 400.

Contractionary Monetary Policy



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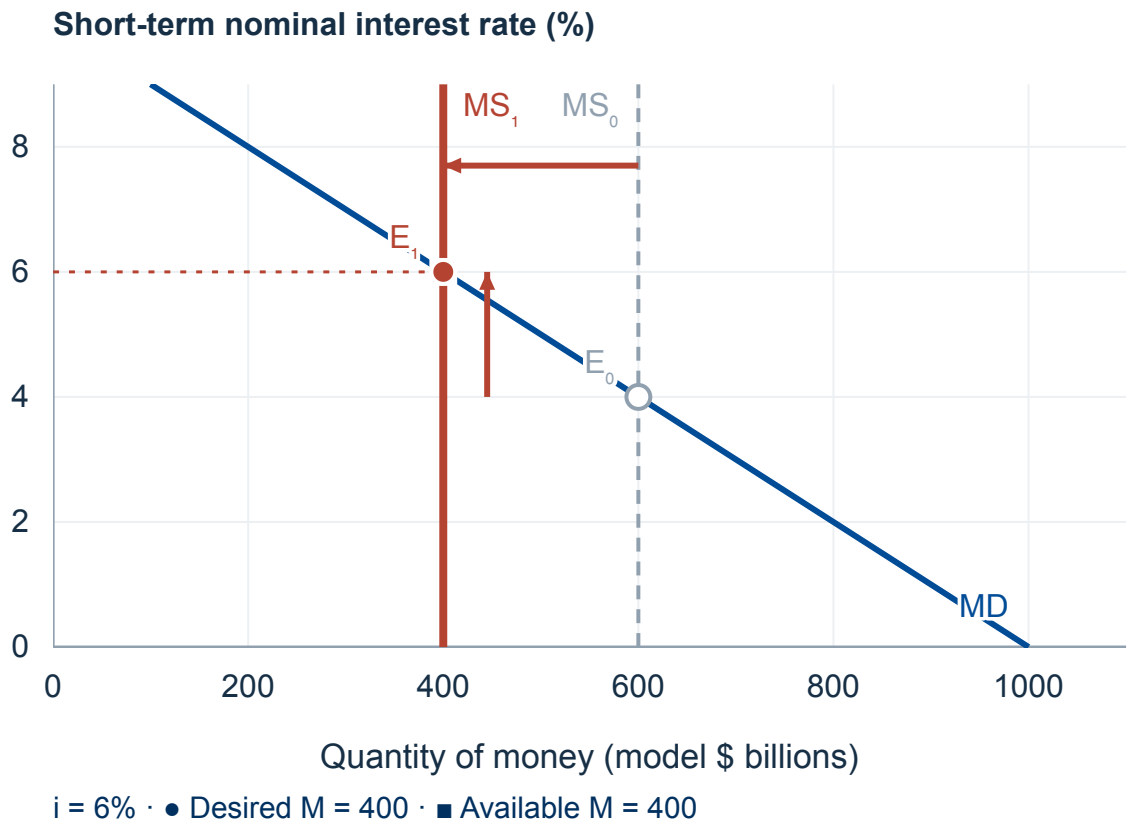
Less money

In this model, a Fed sale lowers money supply to 400.

Bond sales

At 4%, people want more money. They sell bonds to obtain it.

Contractionary Monetary Policy



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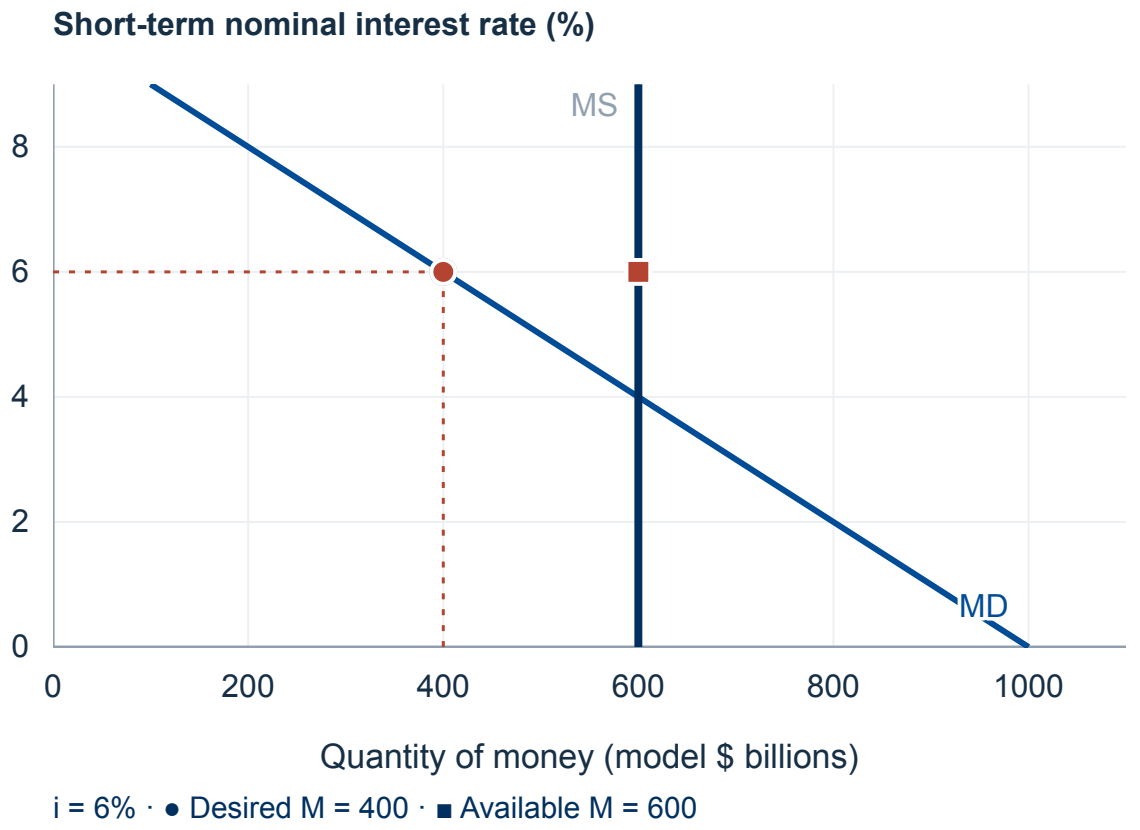
Bond sales

At 4%, people want more money. They sell bonds to obtain it.

Higher interest rate

Bond prices fall. The interest rate rises to 6%.

An Interest Rate above Equilibrium

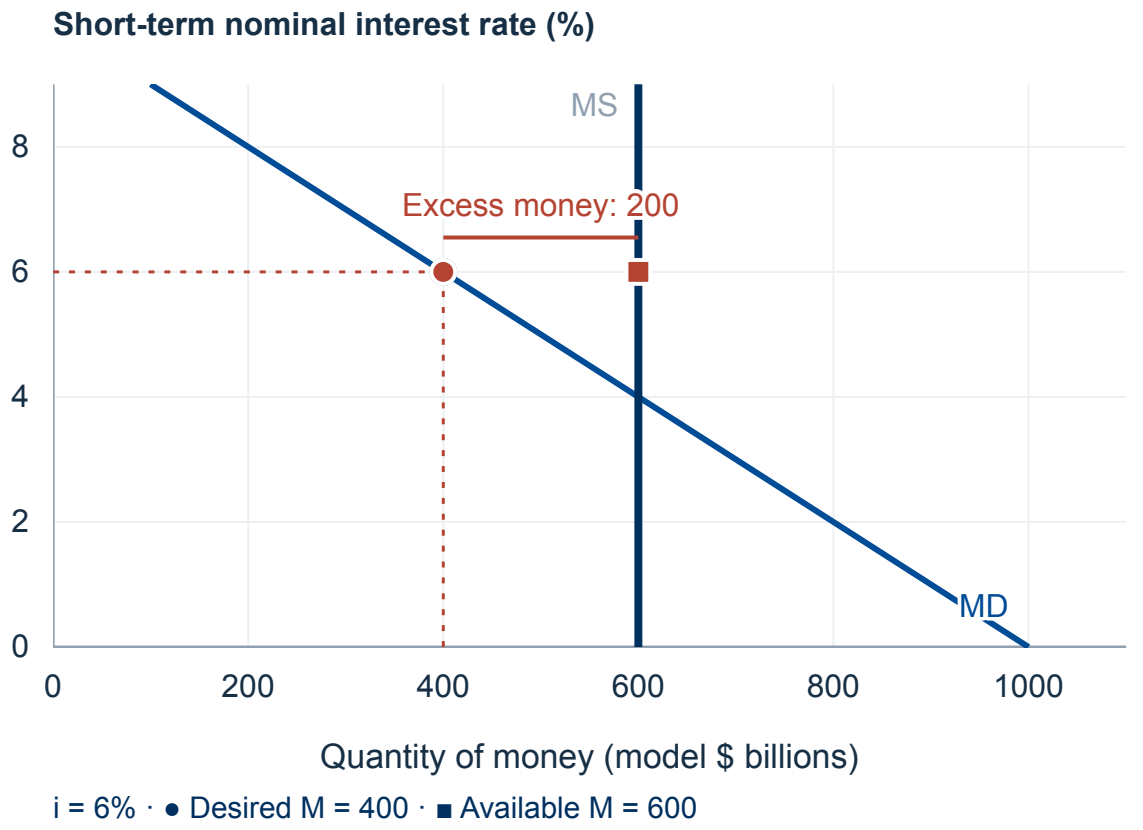


Hold prices, real GDP and the return on money fixed.

At 6%

At 6%, people want 400 in money, but 600 is available.

An Interest Rate above Equilibrium



Hold prices, real GDP and the return on money fixed.

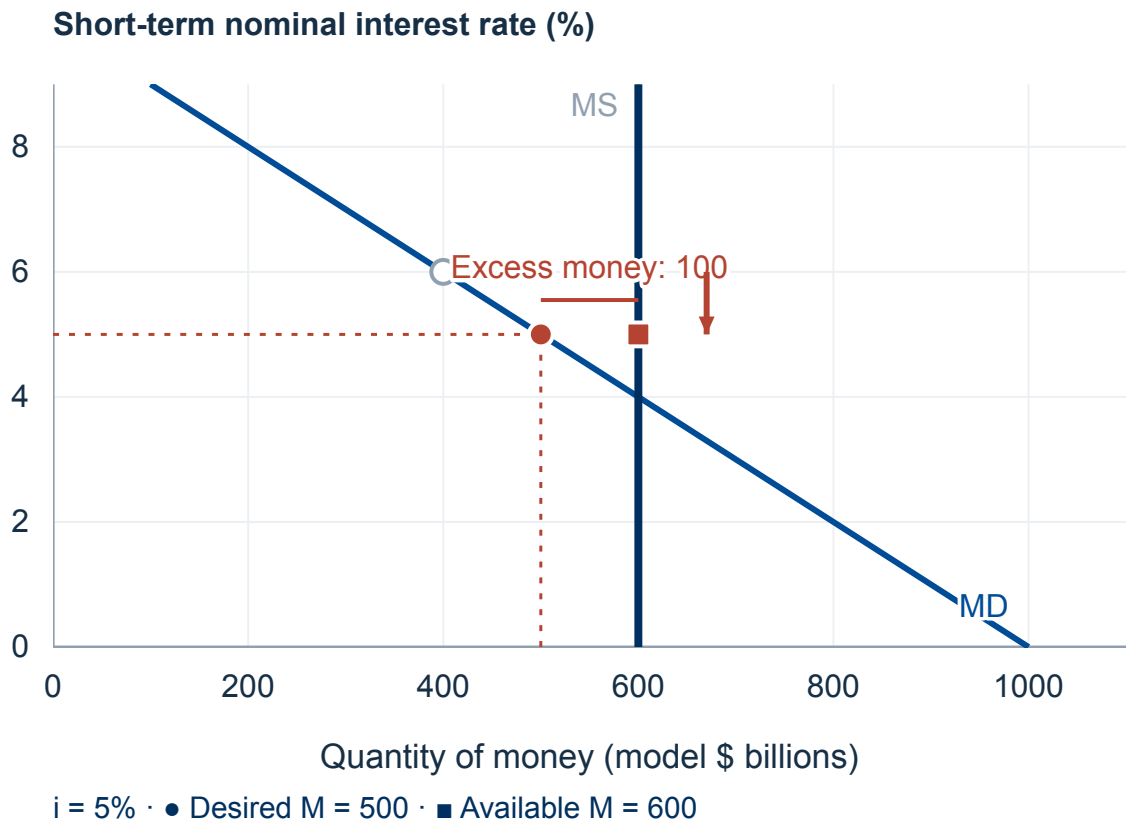
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Excess balances

They use excess money balances to buy bonds.

An Interest Rate above Equilibrium



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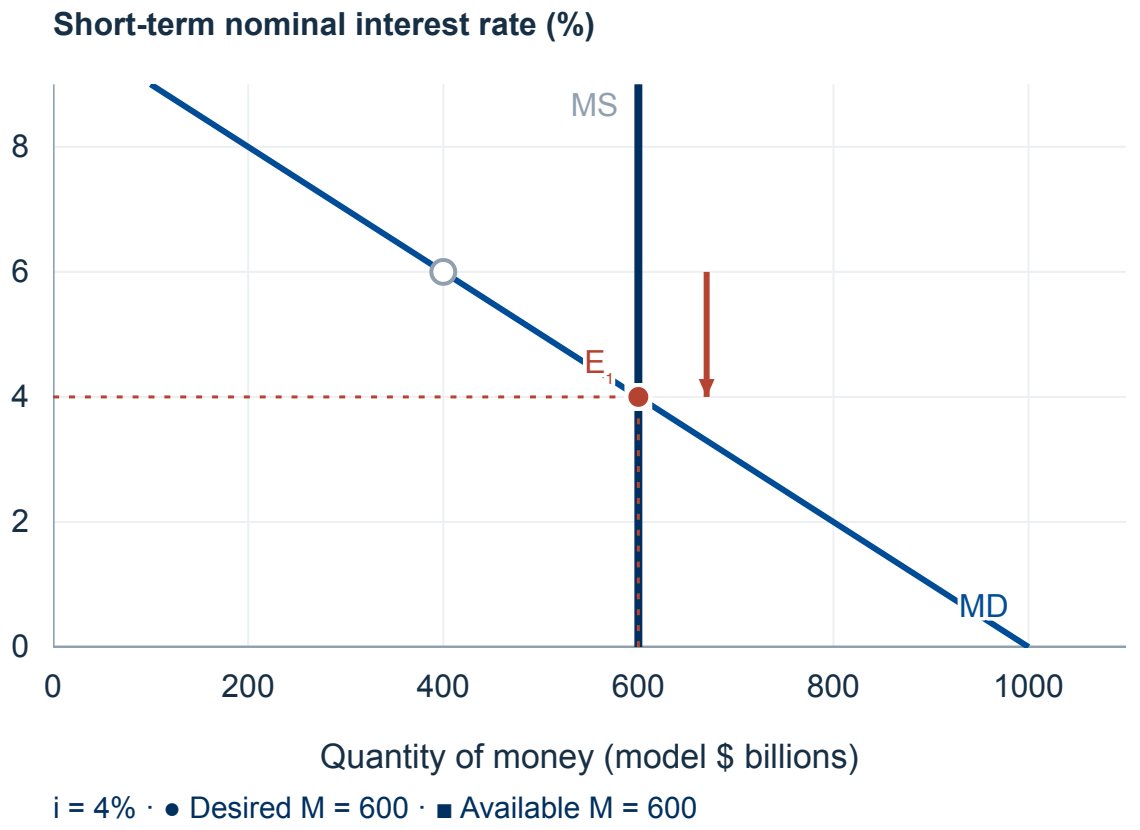
Excess balances

They use excess money balances to buy bonds.

Bond prices and yields

A bond paying $100 next year rises from $94.34 to $96.15. Its yield falls.

An Interest Rate above Equilibrium



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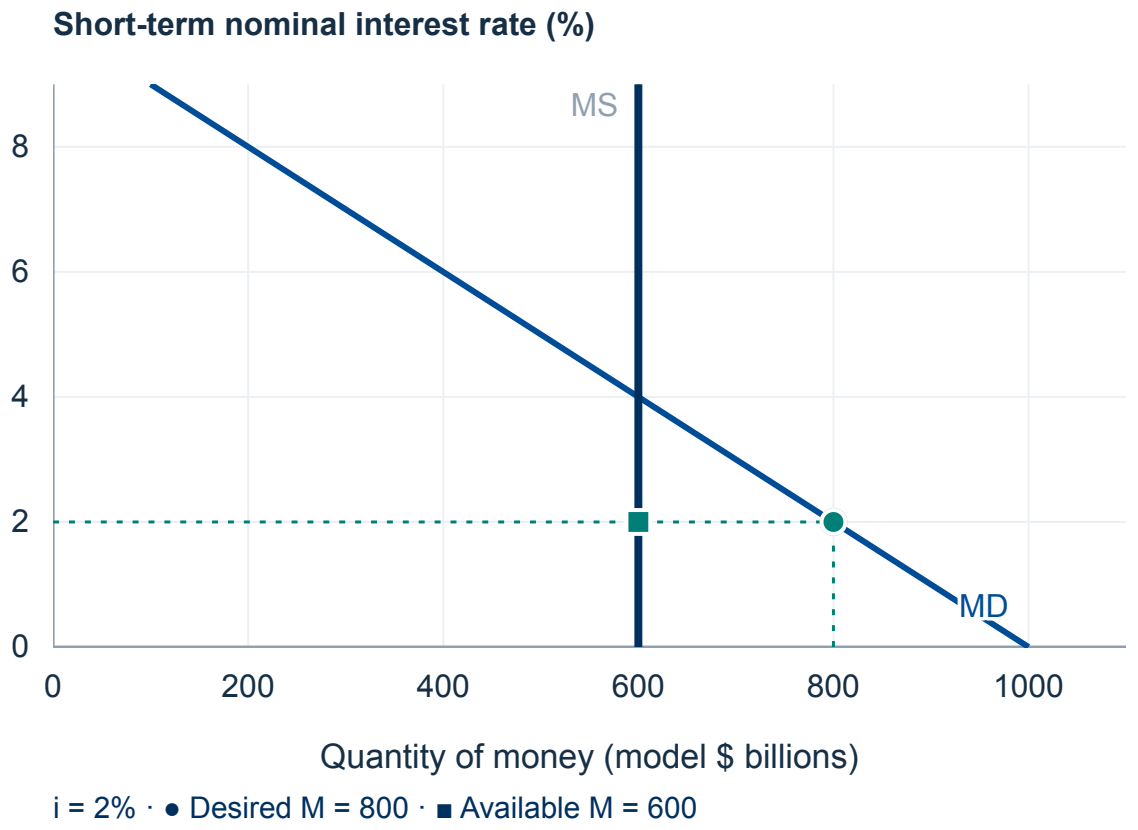
Bond prices and yields

A bond paying $100 next year rises from $94.34 to $96.15. Its yield falls.

At equilibrium

At 4%, money demand and money supply both equal 600.

An Interest Rate below Equilibrium

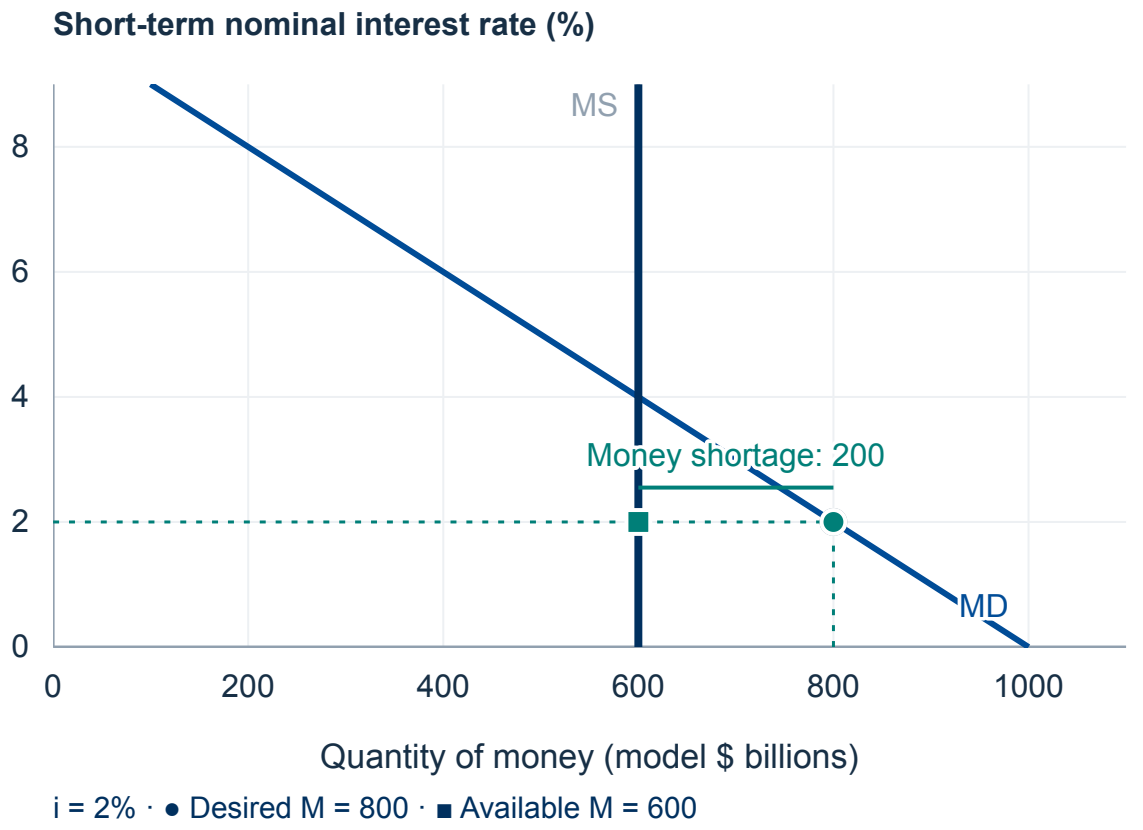


Hold prices, real GDP and the return on money fixed.

At 2%

At 2%, people want 800 in money, but only 600 is available.

An Interest Rate below Equilibrium



Hold prices, real GDP and the return on money fixed.

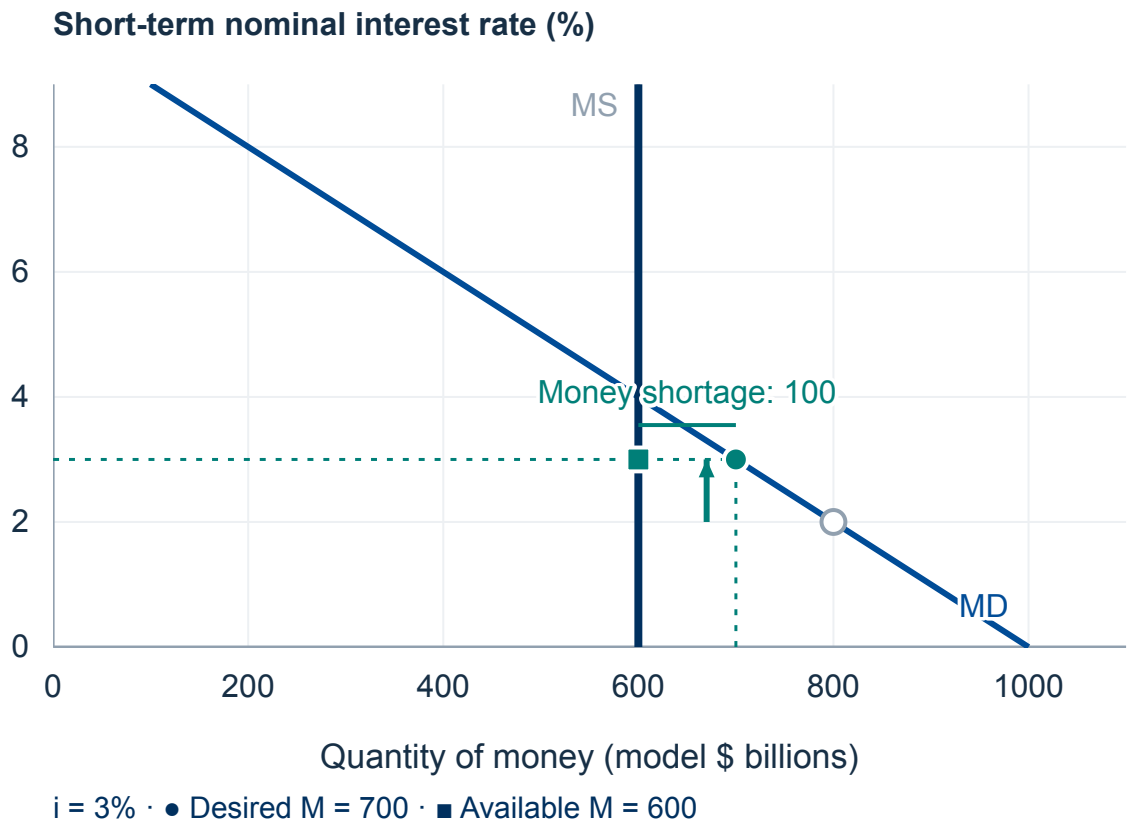
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At 2%, people want 800 in money, but only 600 is available.

A money shortage

They sell bonds to obtain money for payments.

An Interest Rate below Equilibrium



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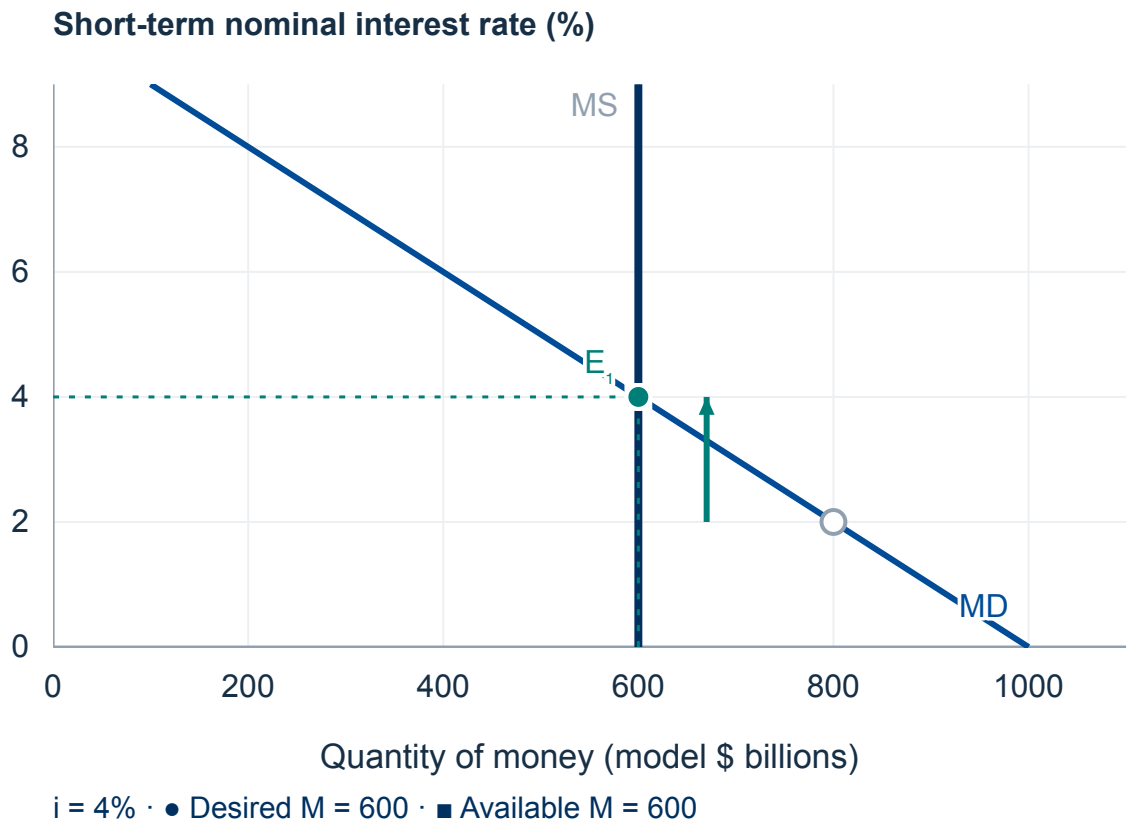
A money shortage

They sell bonds to obtain money for payments.

Bond prices and yields

A bond paying $100 next year falls from $98.04 to $96.15. Its yield rises.

An Interest Rate below Equilibrium



Hold prices, real GDP and the return on money fixed.

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At 2%, people want 800 in money, but only 600 is available.

A money shortage

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Bond prices and yields

A bond paying $100 next year falls from $98.04 to $96.15. Its yield rises.

At equilibrium

At 4%, money demand and money supply both equal 600.

Money Supply and Interest Rates

Policy in the model	Money supply	Bond prices	Short-term rate
Fed buys bonds	Rises	Rise	Falls
Fed sells bonds	Falls	Fall	Rises

Hold prices, real GDP and the return on money fixed.

Money-Market Checkpoint

1. Money supply rises while MD stays fixed. What happens to the interest rate, and why?

2. The rate is below equilibrium. How does the money shortage disappear?

Money-Market Checkpoint

1. Money supply rises while MD stays fixed. What happens to the interest rate, and why?

> MS shifts right. People buy bonds with excess money, raising bond prices and lowering the rate.

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Money-Market Checkpoint

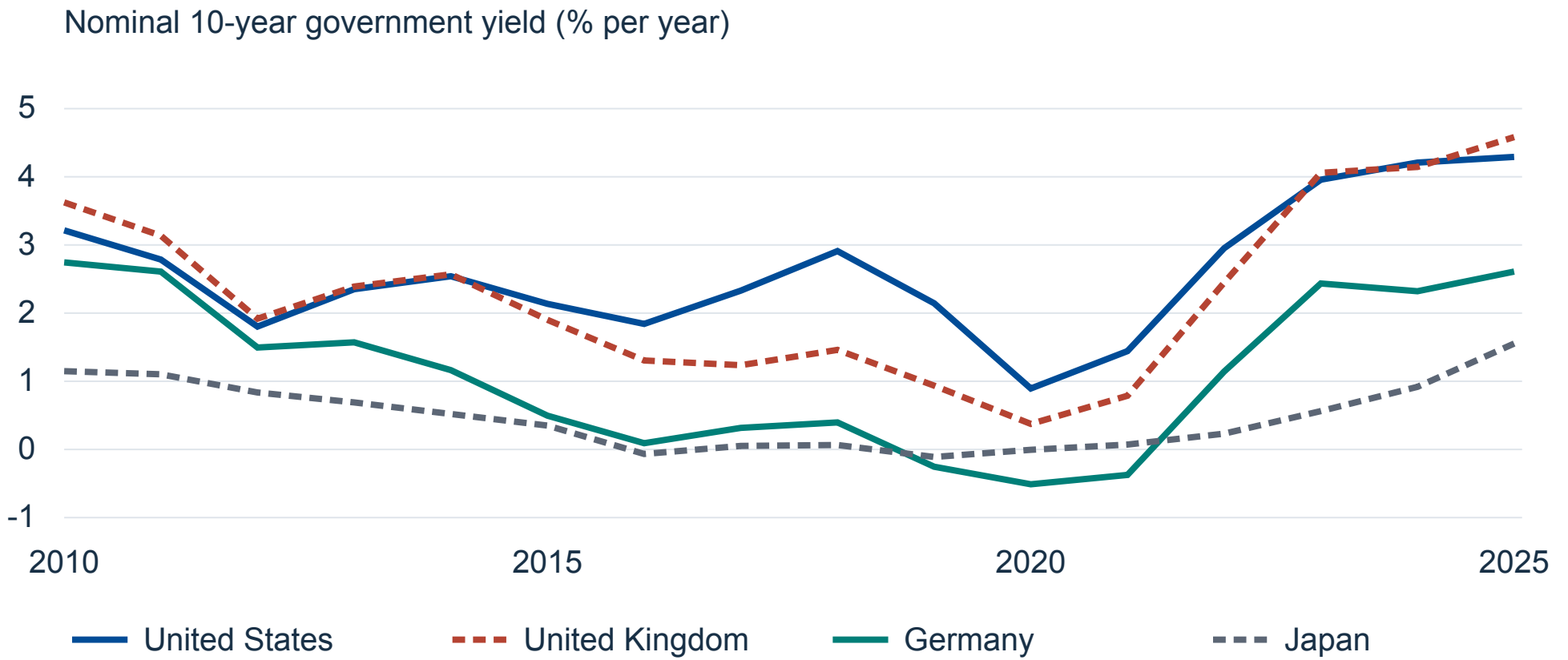
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> MS shifts right. People buy bonds with excess money, raising bond prices and lowering the rate.

2. The rate is below equilibrium. How does the money shortage disappear?

> People sell bonds. Bond prices fall and the rate rises, reducing the quantity of money demanded.

Long-Term Interest Rates across Countries



Annual averages, 2010–2025; own-currency yields. National benchmark definitions differ.

OECD (2026), Financial market statistics: long-term interest rates (accessed November 2, 2026) · OECD data terms, section 3

One-Year and Two-Year Investments

Invest $600 for two years	Year 1 rate	Year 2 rate
Two-year contract, annual compounding	5% fixed	5% fixed
Roll one-year investments: P5	4%	2% scenario
Roll one-year investments: P6	4%	8% scenario

- Answer P5 and P6 before revealing the payoffs.

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- Answer P5 and P6 before revealing the payoffs.
- P5: two years at 5% gives $661.50; rolling at 4% then 2% gives $636.48.

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- Answer P5 and P6 before revealing the payoffs.
- P5: two years at 5% gives $661.50; rolling at 4% then 2% gives $636.48.
- P6: rolling at 4% then 8% gives $673.92, more than the fixed two-year contract.

Longer-Term Interest Rates

- A longer-term rate reflects expected future short-term rates and a premium for longer-term risk.

A longer bond maturity is different from the economy’s long run.

SEC Investor.gov: bonds · Federal Reserve Board: policy goals and transmission

Longer-Term Interest Rates

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SEC Investor.gov: bonds · Federal Reserve Board: policy goals and transmission

Longer-Term Interest Rates

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- An expected Fed rate change may already be reflected in bond prices.
- A cut in the overnight rate does not guarantee an equal fall in a 10-year rate.

A longer bond maturity is different from the economy’s long run.

SEC Investor.gov: bonds · Federal Reserve Board: policy goals and transmission

15.3 Monetary Policy and Aggregate Demand

- Interpret shifts in aggregate demand and demonstrate the short-run effects of monetary policy.

Fiscal and Monetary Policy

Policy	Main tools	Useful for
Monetary policy	Interest rates and financial conditions	Changes in spending across the economy
Fiscal policy	Purchases, taxes and transfers	Direct spending or support for particular groups
Both together	A coordinated response	Severe downturns, especially when rates cannot fall much further

- New fiscal measures may need a law. Automatic stabilizers do not.

IMF: fiscal policy · Federal Reserve Board: policy goals and transmission

Fiscal and Monetary Policy

Policy	Main tools	Useful for
Monetary policy	Interest rates and financial conditions	Changes in spending across the economy
Fiscal policy	Purchases, taxes and transfers	Direct spending or support for particular groups
Both together	A coordinated response	Severe downturns, especially when rates cannot fall much further

- New fiscal measures may need a law. Automatic stabilizers do not.
- Adding demand does not repair a loss of productive capacity. Supply shocks create a policy tradeoff.

IMF: fiscal policy · Federal Reserve Board: policy goals and transmission

Borrowing Costs and Investment

1 A bakery considers a $10,000 oven. Expected net receipts next year are $10,500.

These are business-loan rates, not the federal funds rate.

Federal Reserve Board: policy goals and transmission

Borrowing Costs and Investment

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2. Borrow at 6%: repay $10,600. The project loses $100.

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Federal Reserve Board: policy goals and transmission

Borrowing Costs and Investment

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Federal Reserve Board: policy goals and transmission

Borrowing Costs and Investment

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2. Borrow at 6%: repay $10,600. The project loses $100.
3. Borrow at 4%: repay $10,400. The project gains $100.
4. Cheaper borrowing can raise investment and consumption. AD shifts right.

These are business-loan rates, not the federal funds rate.

Federal Reserve Board: policy goals and transmission

Aggregate Demand Checkpoint

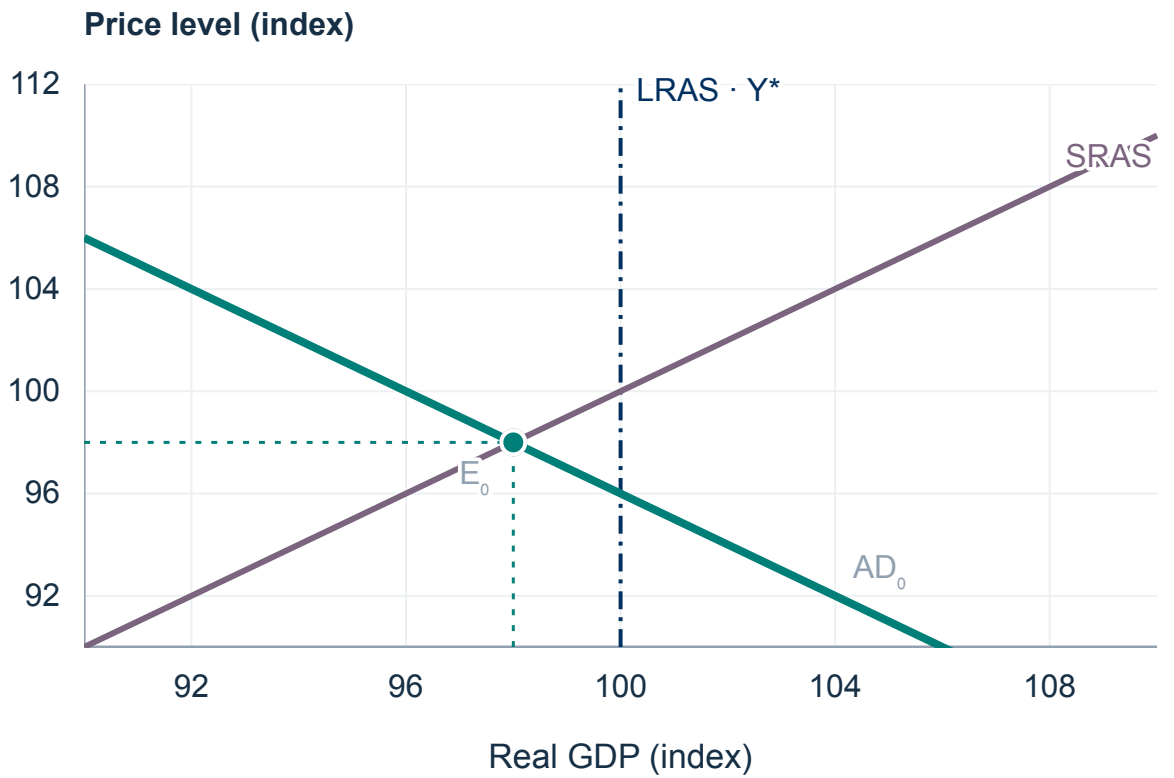
Output is below potential. Lower borrowing costs raise C and I. With SRAS fixed, predict AD, output and the price level.

Aggregate Demand Checkpoint

Output is below potential. Lower borrowing costs raise C and I. With SRAS fixed, predict AD, output and the price level.

> AD shifts right. Short-run output and the price level rise.

Expansionary Policy and Aggregate Demand



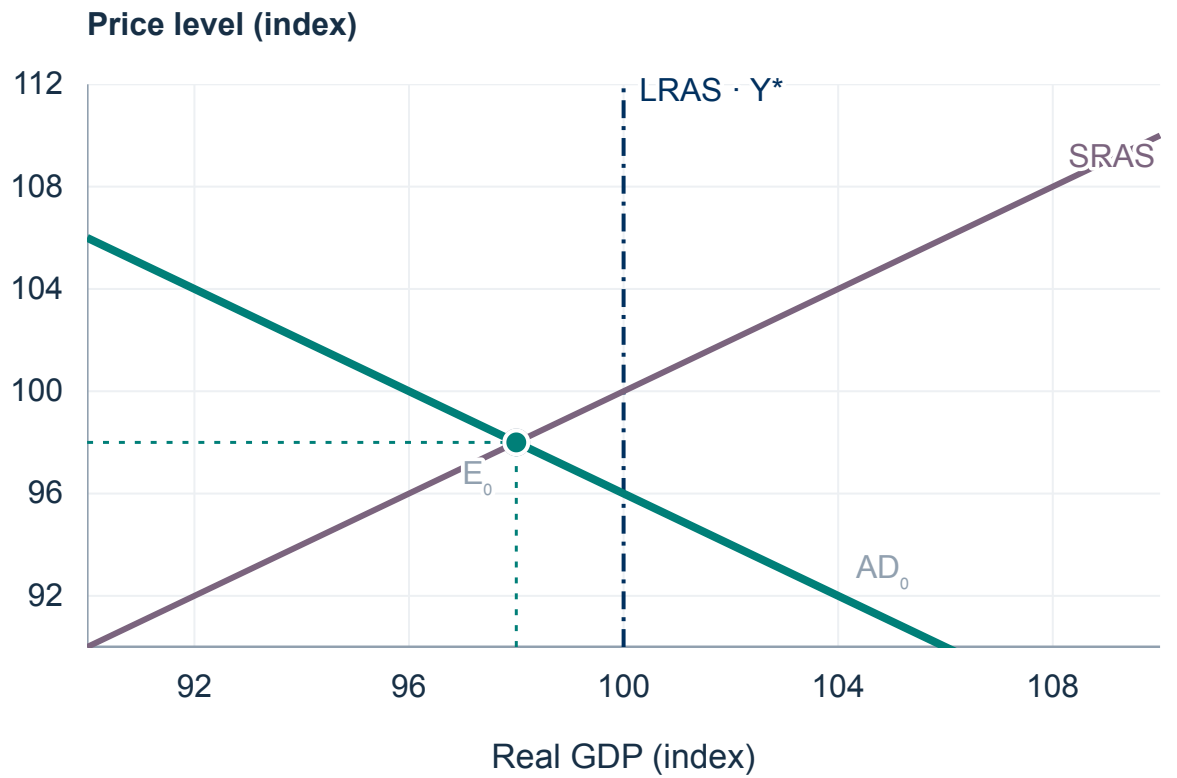
Y = 98 · P = 98 · SRAS and potential output fixed

SRAS and potential output stay fixed.

Below potential

Output is 98; potential output is 100.

Expansionary Policy and Aggregate Demand



Y = 98 · P = 98 · SRAS and potential output fixed

SRAS and potential output stay fixed.

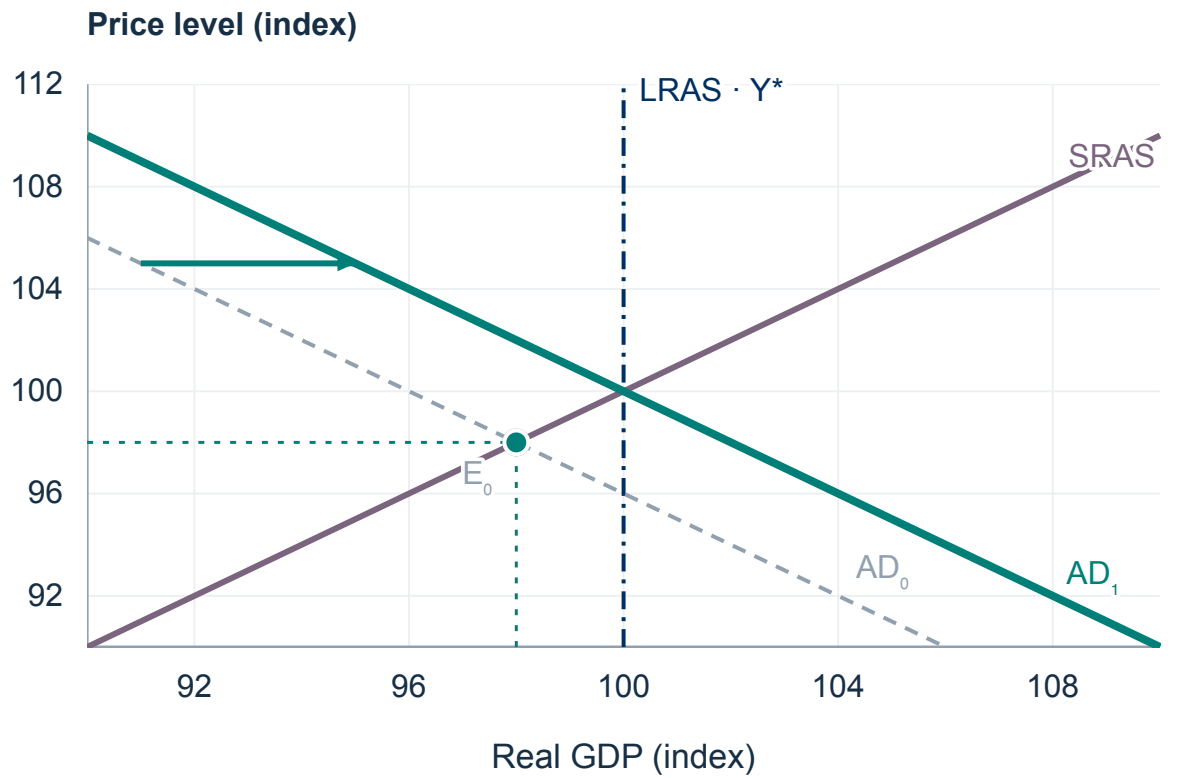
Below potential

Output is 98; potential output is 100.

Spending rises

Lower borrowing costs encourage C and I.

Expansionary Policy and Aggregate Demand



Y = 98 · P = 98 · SRAS and potential output fixed

SRAS and potential output stay fixed.

Below potential

Output is 98; potential output is 100.

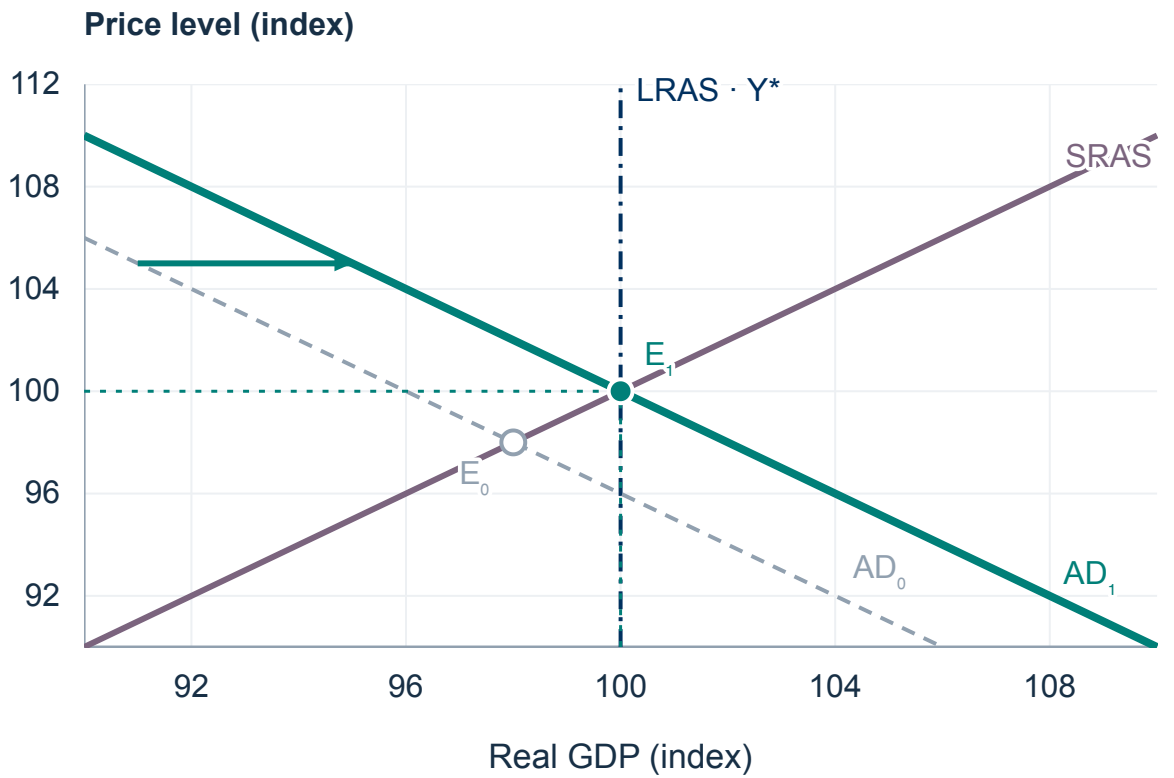
Spending rises

Lower borrowing costs encourage C and I.

AD shifts right

AD shifts right. SRAS and potential output stay fixed.

Expansionary Policy and Aggregate Demand



Y = 100 · P = 100 · SRAS and potential output fixed

SRAS and potential output stay fixed.

Below potential

Output is 98; potential output is 100.

Spending rises

Lower borrowing costs encourage C and I.

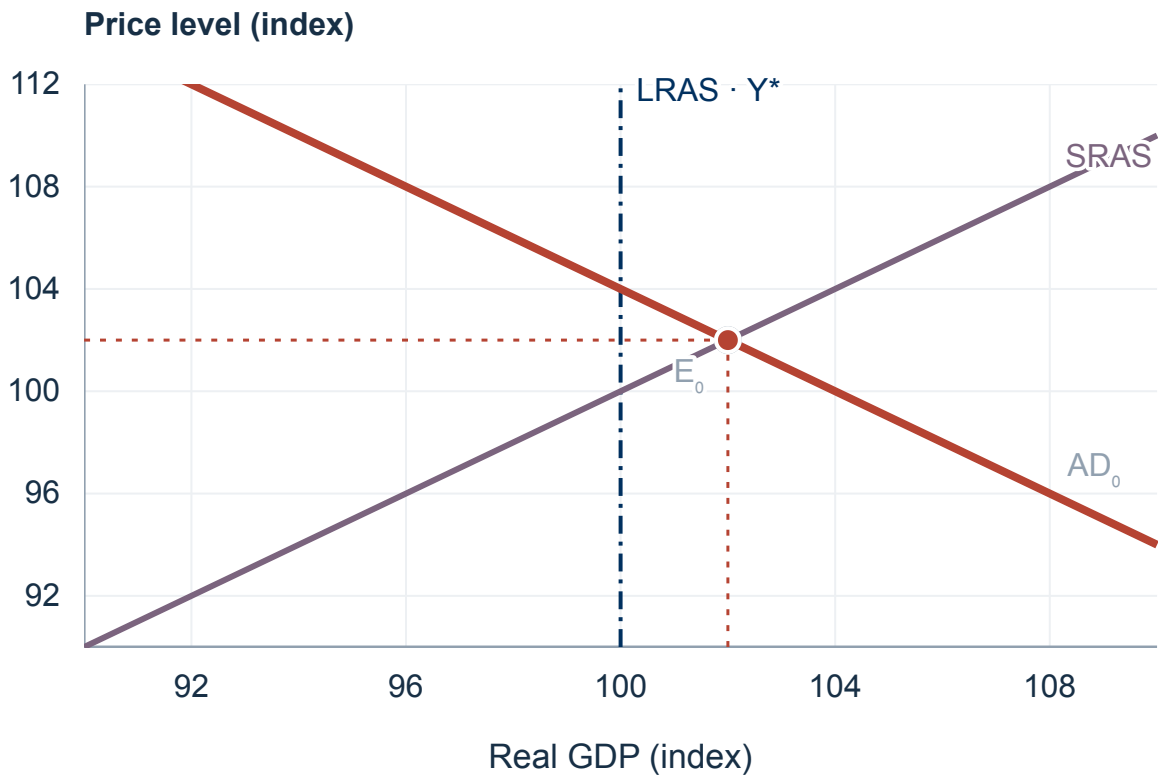
AD shifts right

AD shifts right. SRAS and potential output stay fixed.

New equilibrium

The new equilibrium has output 100 and price level 100.

Contractionary Policy and Aggregate Demand



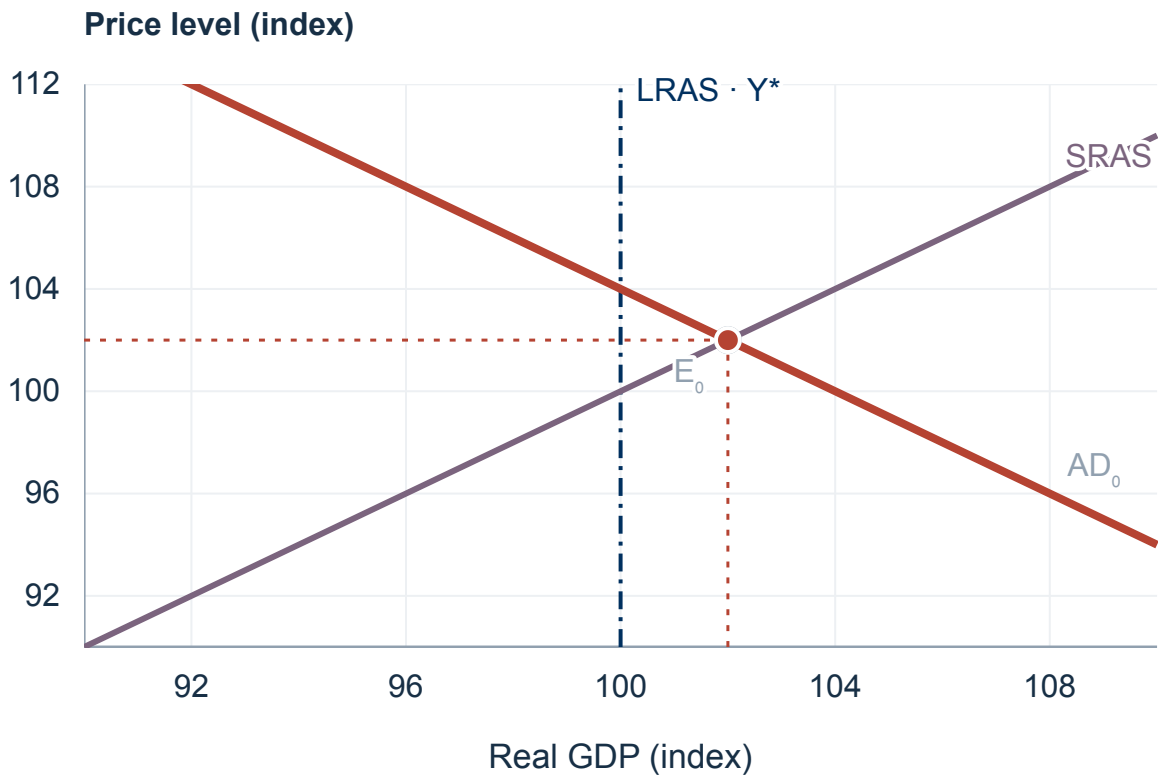
Y = 102 · P = 102 · SRAS and potential output fixed

SRAS and potential output stay fixed.

Above potential

Output is 102; potential output is 100.

Contractionary Policy and Aggregate Demand



Y = 102 · P = 102 · SRAS and potential output fixed

SRAS and potential output stay fixed.

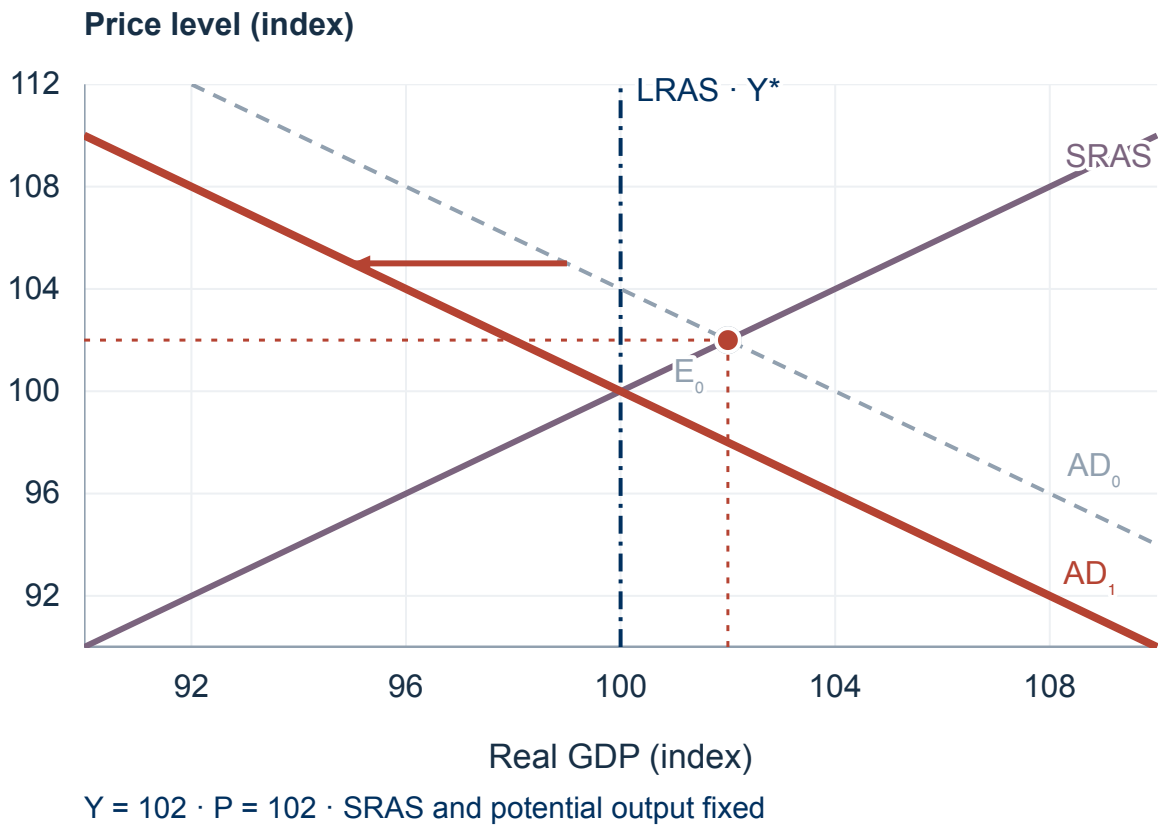
Above potential

Output is 102; potential output is 100.

Spending falls

Higher borrowing costs reduce C and I.

Contractionary Policy and Aggregate Demand



Y = 102 · P = 102 · SRAS and potential output fixed

SRAS and potential output stay fixed.

Above potential

Output is 102; potential output is 100.

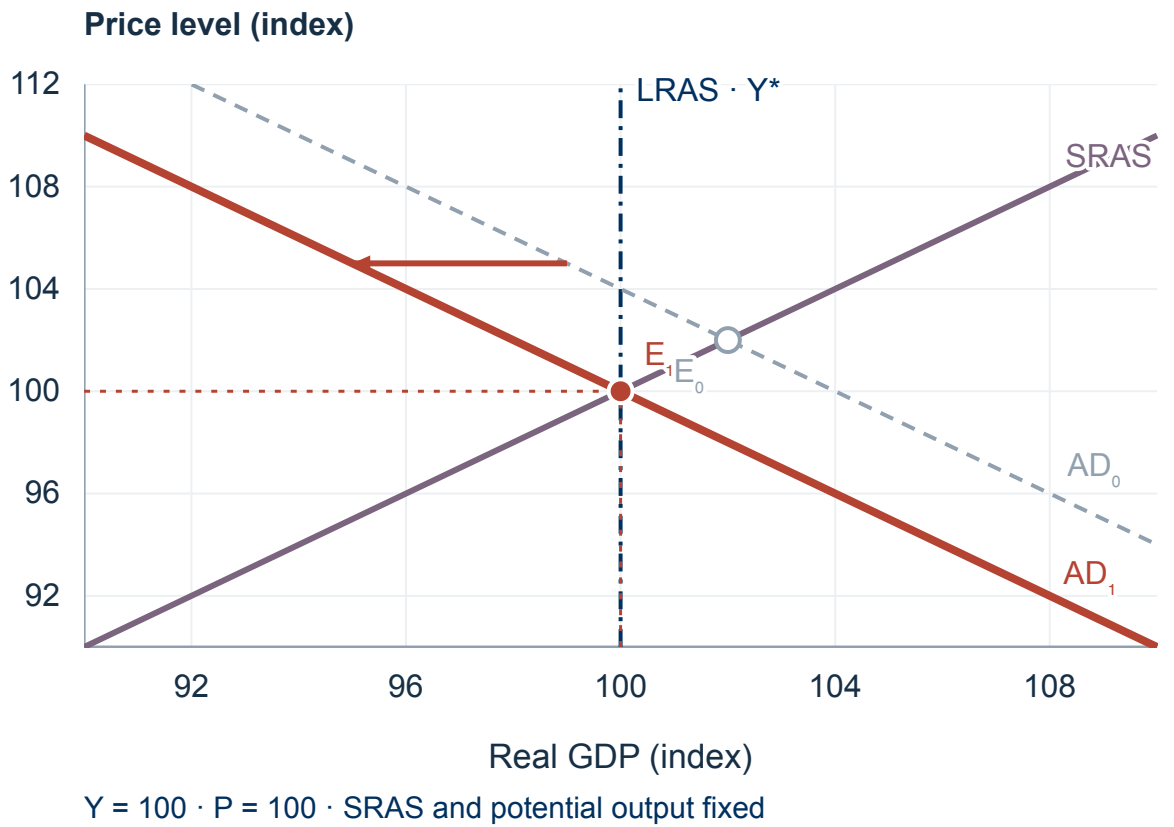
Spending falls

Higher borrowing costs reduce C and I.

AD shifts left

AD shifts left. SRAS and potential output stay fixed.

Contractionary Policy and Aggregate Demand



SRAS and potential output stay fixed.

Above potential

Output is 102; potential output is 100.

Spending falls

Higher borrowing costs reduce C and I.

AD shifts left

AD shifts left. SRAS and potential output stay fixed.

New equilibrium

Output returns to 100. The price level is lower than without the policy.

Monetary Policy: Short-Run Effects

Channel	Expansionary policy	Contractionary policy
Money-market model	MS right; interest rate falls	MS left; interest rate rises
Borrowing costs	Can fall	Can rise
Spending	C and I rise; AD right	C and I fall; AD left
With SRAS fixed	Output and price level rise	Output and price level are lower

Potential output stays fixed in these short-run comparisons.